

FISH

Fish are the most diverse chordates and can be found in all types of watery habitat – from tiny, freshwater pools to the deep ocean. Almost without exception, they breathe using feathery gills to absorb oxygen from the water, and almost all swim using fins.

PHYLUM	CHORDATA
CLASSES	PETROMYZONTI MYXINI ELASMOBRANCHII HOLOCEPHALI ACTINOPTERYGII COELACANTHI DIPNEUSTI
ORDERS	66–81
FAMILIES	560–588
SPECIES	About 33,900



Overlapping scales cover the body of a bicolor parrotfish, forming a flexible layer that protects against injury.



Predatory sharks lunge into a huge shoal of sardines, which swim ever closer together for protection.



The huge mouth of a male gold-speck jawfish provides an unlikely nest for its eggs. He will not feed during incubation.

Fish are not a natural, single group but are in fact seven classes of vertebrate chordates, of which the familiar ray-finned (bony) fish are by far the most populous. The majority of fish are cold-blooded and so their body temperature matches that of the surrounding water. A few top predators, such as white sharks, can maintain a flow of warm blood to the brain, eyes, and main muscles, which allows them to hunt actively even in very cold water. Most fish are protected by scales or bony plates embedded in the skin. In fast swimmers these are lightweight, and provide streamlining as well as protection from abrasion and disease. Although a few fish slither or shuffle over the seabed, most swim using fins. With the notable exception of rays, the tail fin usually provides the main propulsive force. Paired pectoral and pelvic fins (on the sides and underside respectively) provide stability and manoeuvrability, helped by up to three dorsal fins on the fish's upper side and one or two anal fins. Fish avoid collisions – especially in large shoals – by using special sensory organs that detect the vibrations made as they or other animals move through the water. Most fish have a row of these organs running along both sides of the body, called the lateral line. While they also use the senses of hearing, touch, sight, taste, and smell found in other chordates, only fish have a lateral line system.

REPRODUCTIVE STRATEGIES

The seven classes of fish have very different ways of reproducing. Most ray-finned fish have external fertilization and shed masses of eggs and sperm directly into the water to compensate for the huge numbers that are eaten before they can develop into juvenile fish; a few instead guard their eggs until hatching. In contrast, cartilaginous fish (sharks, rays, and chimaeras) all have internal fertilization and produce eggs or young at an advanced developmental stage. This requires a high input of energy – only a few young are produced at one time, but these have a good chance of survival. The eggs of lampreys (jawless fish) hatch into larvae that live and feed for many months before undergoing metamorphosis into the adult form.

BRIGHT SPECTACLE >

The fabulous colours of the Banggai cardinalfish, seen here sheltering among giant anemones, attract aquarium owners.

